Rectifiers, Metallic, Selenium (High Temperature)

1. SCOPE

1.1 This specification covers selenium metallic rectifiers consisting of one or more cells, for operation up to 125°C. (See 6.1.)

1.2 Type designation. - The type designation shall be in the following form:

RS15	B	101	BS	122
Style (1.2.1)	Mounting (1.2.2)	DC Voltage rating (1.2.3)	Circuit configuration (1.2.4)	DC Current rating (1.2.5)

1.2.1 Style. - The style is identified by the 2-letter symbol "RS" followed by a 2-digit number; the letters identify selenium metallic rectifiers and the number identifies the rectifier cell size in accordance with Table I.

Table I - Style

Style	Cell Size	Style	Cell Size
RS15	1 x 1	RS55	3 x 3
RS25	1.25 x 1.25	RS65	4 x 4
RS35	1.6 x 1.6	RS75	4 x 6
RS45	2.2 x 2.2	RS85	5 x 6

1.2.2 Mounting. - The method of mounting is identified by a single letter in accordance with Table II.

Table II - Mounting method

Symbol	Method
B	Bolt
S	Stud

1.2.3 DC voltage rating. - The nominal dc output voltage rating when the rectifier is connected to a resistive load is identified by a 3-digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow. The value designations are as shown in Table III.

Table III - Designation of dc voltage rating values

Designation	DC voltage rating (80°C ambient temp.)
100 to 800	10 to 80 volts
101 to 401	100 to 400 volts

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1.2.4 Circuit configuration. - Circuit configuration is identified by two letters in accordance with Table IV. Circuit diagrams shall be as shown on Figure 1.

Table IV - Circuit configuration

Symbol	Circuit configuration
HS	Half wave, single phase
CS	Center tap, single phase
BS	Bridge, single phase
DS	Doubler, single phase

1.2.5 DC Current rating. - The nominal dc output current rating when the rectifier is connected to a resistive load is identified by a 3-digit number; the first two digits represent significant figures in milliamperes and the last digit specifies the number of zeros to follow. The value designations are as shown in Table V.

Table V - Designation of dc current rating values

Designation	DC Current rating (80°C ambient temp.)
600 to 800	.060 to .080 amperes
101 to 901	.100 to .900 amperes
102 to 902	1 to 9 amperes
103 to 903	10 to 90 amperes

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and other publications of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-621	Boxes; wood, nailed and lock-corner
QQ-S-571	Solder, soft (tin, tin-lead, and lead silver)
TT-C-595	Colors; (for) ready-mixed paints
UU-T-111	Tape; paper, gummed (sealing and securing)
LLL-F-291	Fiberboard; corrugated, single face, (flexible)
PPP-B-566	Boxes, folding, paper board

MILITARY

MIL-P-14	Plastic-materials, molding, and Plastic-parts, molded; thermosetting
MIL-P-79	Plastic-materials, laminated, thermosetting rods and tubes
JAN-P-106	Packaging and packing for overseas shipment - boxes; wood, nailed (for weight of contents not in excess of 1,000 pounds)

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JAN-P-108	Packaging and packing for overseas shipment-boxes, fiberboard (V-board and W-board) exterior and interior.
MIL-P-116	Preservation, methods of.
JAN-P-127	Packaging and packing for overseas shipment - tape, adhesive, pressure-sensitive, water resistant.
MIL-B-131	Barrier material, water vapor proof, flexible.
MIL-S-901	Shockproof equipment, class HI (high-impact), shipload.
MIL-S-933	Screws, machine, cap and set; and nuts.
MIL-T-10513	Tropicalization of Signal Corps equipment.
MIL-M-13231	Marking of electronic items.
MIL-P-15037	Plastic-material, laminated thermosetting, sheets, glass-cloth, melamine-resin.

STANDARDS

MILITARY

MIL-STD-105 with Appendix	Sampling procedures and tables for inspection by attributes.
MIL-STD-109	Inspection terms and definitions.
MIL-STD-129	Marking for shipment and storage.
MIL-STD-202	Test methods for electronic and electric component parts.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publication. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.).

ASSOCIATION OF AMERICAN RAILWAYS

Uniform Freight Classification Ratings, Rules and Regulations.

(Copies of Uniform Freight Classification Ratings, Rules and Regulations may be obtained from Association of American Railways, Chicago, Illinois.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification Rules and Container Specifications for Truck Shipment.

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(Copies of National Motor Freight Classification Rules and Container Specifications for Truck Shipments may be obtained from the American Trucking Association, 1424 Sixteenth Street, N. W., Washington 6, D. C.).

3. REQUIREMENTS

3.1 Definitions. -

3.1.1 Rectifying element. - A rectifying element is a circuit element which conducts current effectively in one direction only.

3.1.2 Rectifying junction. - A rectifying junction is the region which exhibits the asymmetrical conductivity in a rectifier.

3.1.3 Rectifier. - A rectifier is a single structure of one or more rectifier cells which utilizes the asymmetrical conductivity of the junction between dissimilar solid conducting materials to obtain a current flow effectively in one direction only.

3.1.4 Rectifier cell. - A rectifier cell is an elementary rectifying device having one positive electrode, one negative electrode, and one rectifying junction.

3.1.5 Rectifier assembly. - A rectifier assembly is a combination of one or more rectifiers including all of the rectifying elements used in one rectifier circuit.

3.1.6 Forward direction. - Forward direction is the direction of lesser resistance to current flow through a rectifier cell, i.e., from the negative electrode to the positive electrode.

3.1.7 Forward current. - Forward current is the current that flows through a rectifier cell in the forward direction.

3.1.8 Forward voltage drop. - Forward voltage drop is the voltage drop which results from the flow of current through a rectifier cell in the forward direction.

3.1.9 Reverse direction. - Reverse direction is the direction of greater resistance to current flow through a rectifier cell, i.e., from the positive electrode to the negative electrode.

3.1.10 Reverse voltage. - Reverse voltage is the voltage which is applied to a rectifier cell in the reverse direction.

3.1.11 Reverse current. - Reverse current is the current which flows through a rectifier cell in the reverse direction.

3.1.12 Applied voltage rating (rms). - The rms applied voltage rating of a rectifier cell is the maximum sinusoidal rms voltage which can be applied to the cell under specified conditions.

3.1.13 Basic rating. - The basic rating of a rectifier is the dc output current and ac applied voltage at which the rectifier is rated for continuous operation at specified load with convection cooling in a + 80°C ambient temperature.

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3.1.14 Half-wave rectifier. - A half-wave rectifier is a rectifier which changes alternating current into pulsating unidirectional current, utilizing only one-half of each cycle.

3.1.15 Full-wave rectifier. - A full-wave rectifier is a rectifier which changes alternating current into pulsating unidirectional current, utilizing both halves of each cycle.

3.1.16 Basic bridge rectifier. - A basic bridge rectifier is a single phase, full wave rectifier made-up of four elements each consisting of a single rectifier cell.

3.1.17 Ambient temperature. - Ambient temperature is the temperature of the medium into which the heat of the rectifier is dissipated.

3.2 Preproduction samples. -

3.2.1 Quantities. - Sixteen (16) rectifiers of each type on order shall be furnished for Government preproduction inspection and approval.

3.2.2 Fabrication and assembly. - Preproduction samples shall be fabricated by the contractor and shall meet all requirements specified for the rectifiers on contract. The preproduction samples shall be assembled using materials and processes that will be employed in production, and shall be fabricated and assembled in a manner similar to that used in production.

3.2.3 Statement. - Preproduction samples shall be accompanied by a statement comprised of the following:

(a) Certification that the preproduction samples have been completely fabricated using materials and processes identical to those to be used in production; or (1) and (2) below, as applicable.

(1) Detailed description of each point of difference between the techniques, tools, methods, etc. used in fabricating and assembling the samples and those to be used in production.

(2) Detailed description of any point of difference between the materials used in the samples and those to be used in production, and reason why the difference was unavoidable.

(b) Identity of the plant where the samples were fabricated and date when completed.

(c) Test data showing compliance of the samples with specified performance. The test data shall comprise an engineering report, giving test procedure, observations and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, etc.).

3.2.4 Approval. - Approval of preproduction samples shall not be construed as a waiver of any specified requirement.

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3.3 Parts, materials, and processes; general. -

3.3.1 Selection. - Parts, materials, and processes shall be such as to enable rectifiers to meet their performance and other requirements, as specified herein, and shall be selected in accordance with the following order of precedence: (When none apply, specified performance and other requirements govern exclusively.)

1. Detailed description of the part, material, or process, by text or graphically.
2. Reference to a detailed document (specification, standard, drawing, handbook, manual, etc.) for the particular purpose of specifying the part, material, or process.

3.4 Plastic material. -

3.4.1 Laminated thermosetting plastic material. - All laminated plastic thermosetting material shall be type GMG in accordance with Specification MIL-P-79 and MIL-P-15037 except that the electrical insulating tube over the mounting bolt or stud or other parts may be Type PBE form Tm in accordance with Specification MIL-P-79 provided this insulating tube or other parts are completely and solidly enclosed by other parts of the rectifier or rectifier assembly.

3.4.2 Molded thermosetting plastic material. - All molded thermosetting materials shall be type MA1 in accordance with Specification MIL-P-14.

3.5 Fungus-resistant materials. - As far as practicable, all non-metallic parts of rectifiers shall be fungus resistant. When it is necessary to use materials not fungus resistant, they shall be treated in accordance with Specification MIL-T-10515.

3.6 Screws, bolts, nuts, and threads. - Screws, bolts, nuts, and other threaded parts shall be of corrosion-resistant material or shall be suitably protected against corrosion. All external threaded parts shall be in accordance with Handbook H28.

3.7 Design and construction. - Rectifiers shall be of the design, construction and physical dimensions specified in Figures 2 and 4 and as follows:

3.7.1 Terminals and connections. - Each rectifier shall be completely wired and terminations shall be located so as to be readily accessible. Terminals shall protrude from the top of the rectifier and shall not extend beyond the protracted width of the cells. AC terminals shall be located at the extremities of the rectifier.

3.7.1.1 Terminal binding screws. - For rectifiers consisting of cells with overall areas larger than 9 square inches, terminal binding screws shall be provided for all terminations for connections external to the rectifier. The screws shall be brass with type A (hexagon) nuts in accordance with Specification MIL-S-933. The screw size shall be in accordance with Table VI.

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Table VI - Terminal Sizes

Current ratings (max)	screw size
15 amperes	6 - 32
25 amperes	8 - 32
50 amperes	10 - 32
100 amperes	1/4 - 20

3.7.1.2 Terminals for soldering. - Rectifiers constructed with cells of an overall area of 9 square inches or less, shall have tin-coated terminations suitable for soldering. Terminations shall be such that wires can be hooked, wrapped around, or otherwise mechanically secured to the terminals prior to soldering, in a manner such that solder alone will not be depended upon for mechanical strength.

3.8 Soldering. -

3.8.1 Solder. - Solder used for electrical connections shall be composition Sn60 conforming to Specification QQ-S-571.

3.8.2 Flux and cleaning agents. - No acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall the acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin, rosin and alcohol, or rosin and turpentine.

3.9 Marking. -

3.9.1 Identification of product. - Marking shall conform to Specification MIL-M-13231.

3.9.1.1 Rectifiers with large cells. - Rectifiers constructed with cells larger than 2.2 inches square shall have the following information marked on the rectifier in such a manner that it will be legible at the completion of the required tests: manufacturer's name, or approved code designation, manufacturer's catalogue number, or identification symbols for the rectifier, type designation, and date of manufacture, using the date code in accordance with Table VII.

Table VII - Date Code /1

January 1950 through December 1959				January 1960 through December 1969			
Month	Month Code	Year Code	Year	Year	Year Code	Month Code	Month
January	A	0	1950	1960	0	A	January
February	B	1	1951	1961	1	B	February
March	C	2	1952	1962	2	C	March
April	D	3	1953	1963	3	D	April
May	E	4	1954	1964	4	E	May
June	F	5	1955	1965	5	F	June
July	G	6	1956	1966	6	G	July
August	H	7	1957	1967	7	H	August
September	J	8	1958	1968	8	J	September
October	K	9	1959	1969	9	K	October
November	L					L	November
December	M					M	December

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1/ The code uses one number and one letter. In the period starting January 1950, the letter is to the left in the code and in the period starting with Ja 1960 the letter is to the right.

3.9.1.2 Rectifiers with small cells. - Rectifiers constructed of cells 2.2 inche square or less shall have the manufacturer's name or approved code designation, an the rectifier type designation marked in such a manner that it will be readable at the completion of all the required tests.

3.9.2 Color coding of terminals. - Positive terminals shall be color coded red; negative terminals shall be color coded black; common terminals which are both positive and negative shall be color coded yellow. Color coding shall be distinct at the completion of all the required tests.

3.10 Forward voltage drop. - The forward maximum voltage drop for rectifier styles RS15, RS25 and RS35 shall be as listed on figures 5 thru 12. Forward voltage drop for styles RS45 thru RS85 shall be the same as RS15 thru RS35 for similar type designations. (See 4.4.2)

3.11 Reverse current. - Maximum reverse current for rectifiers shall be as specified in Table VIII. (See 4.4.3).

Table VIII - Maximum Reverse Current (Milliamperes)

Style	Half Wave	Center tap	Bridge
RS15	5	5	10
RS25	10	10	20
RS35	20	20	40
RS45	30	30	60
RS55	80	80	160
RS65	130	130	260
RS75	180	180	360
RS85	230	230	460

3.12 Dielectric strength. - Rectifiers shall be capable of withstanding the potential specified in the test of 4.4.4 without damage, arcing, or breakdown.

3.13 Insulation resistance. - Unless otherwise specified, rectifiers shall have an insulation resistance of at least 100 megohms. (See 4.4.5)

3.14 Low-temperature exposure. - Following exposure to -65^{+0}_{-3}°C as describe in 4.4.6, rectifiers shall not exceed a 10 percent maximum allowable change in forward voltage drop. Evidence of peeling of the protective finish will be considered a failure but cracking of the protective finish without peeling will not be considered a failure.

3.15 Moisture resistance. - When rectifiers are tested as specified in 4.4.7, t change in forward voltage drop shall not exceed 25 percent, the reverse current sha not exceed 5 percent of the forward current, there shall be no dielectric breakd and the insulation resistance shall be 20 megohms or greater.

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3.16 Low-temperature operation. - Rectifiers shall withstand the low-temperature operation test of 4.4.8 without exceeding 100 percent change in forward voltage drop. The reverse current shall not exceed 5% of the forward current.

3.17 High temperature operation. - When rectifiers are tested as specified in 4.4.9 the reverse current shall not exceed 5% of the forward current.

3.18 Life. - When rectifiers are tested as specified in 4.4.10, the maximum allowable change in forward voltage drop shall be 100%.

3.19 Mechanical shock. - When rectifiers are tested as specified in 4.4.11, the change in forward voltage drop shall not exceed 5 percent, the change in reverse current shall not exceed 5 percent and there shall be no dielectric breakdown.

3.20 Vibration. - When rectifiers are tested as specified in 4.4.12, the change in forward voltage drop shall not exceed 5%, the change in reverse current shall not exceed 5 percent and there shall be no dielectric breakdown. There shall be no looseness of parts or any evidence of mechanical damage.

3.21 Corrosion. - When rectifiers are tested as specified in 4.4.13, the change in forward voltage drop shall not exceed 8 percent, the change in reverse current shall not exceed 8 percent and there shall be no dielectric breakdown. Evidence of peeling of the protective finish will be considered a failure but cracking of the protective finish without peeling will not be considered a failure.

3.22 Workmanship. - The rectifiers shall be manufactured and assembled in a thoroughly workmanlike manner, in accordance with the applicable portions of the following paragraphs herein:

3.4 General requirements for plastic materials and plastic parts.
3.6 Screws, bolts, nuts and threads.
3.8 Flux and cleaning agents (for soldering).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection, general.-

4.1.1 Definitions.- Standard MIL-STD-109 applies for definitions of inspection terms used herein.

4.1.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection.

(b) Acceptance inspection.

4.2 Preproduction inspection.- Preproduction inspection shall be in accordance with table IX.

Table IX Preproduction Tests

Test	Rqt. Par.	Test Par.
Test Group I (All samples)		
Visual and mechanical inspection 1/		4.4.1
Forward voltage drop	3.10	4.4.2
Reverse current	3.11	4.4.3
Dielcctric strength	3.12	4.4.4
Insulation resistance	3.13	4.4.5
Test Group II (4 samples)		
Low-temperature exposure	3.14	4.4.6
Moisture resistance	3.15	4.4.7
Test Group III (4 samples)		
Low-temperature operation	3.16	4.4.8
High-temperature operation	3.17	4.4.9
Test Group IV (4 samples)		
Life	3.18	4.4.10
Test Group V (4 samples)		
Mechanical shock	3.19	4.4.11
Vibration	3.20	4.4.12
Corrosion	3.21	4.4.13

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1/ Marking will be considered defective only if it is illegible at the completion of any of the required tests.

4.3 Acceptance inspection. - Acceptance inspection shall include group A, group B, and group C inspection as specified in 4.3.1 thru 4.3.3. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.3.1 Group A inspection. - Group A inspection (including sampling) shall conform to table X and Standard MIL-STD-105.

Table X - Government Group A Inspection

Inspection	Reqt Par	Test Par	AQL Major	AQL Minor
Visual and Mechanical	3.22	4.4.1		
Dimensions	Fig. 2 and 4	-)		
Parts and Materials	3.3 to 3.6	-)	1.0%	4.0%
Terminals	3.7.1 to 3.7.1.2	-)	Major	Minor
Soldering and cleaning	3.8 to 3.8.2	-)	for	for
Marking	3.9 to 3.9.2	-)	Group	Group
Electrical				
Forward voltage drop	3.10	4.4.2)		
Reverse current	3.11	4.4.3)	1.0%	
Dielectric strength	3.12	4.4.4)	for	*
Insulation resistance	3.13	4.4.5)	Group	

* All electrical defects are considered major.

4.3.2 Group B inspection. - Group B inspection (including sampling) shall conform to table XI with Appendix to Standard MIL-STD-105. Unless otherwise specified herein normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. Disposition of rejected product (sample units and inspection lots) shall be in accordance with Standard MIL-STD-105.

4.3.2.1 Group B sampling plans. - The group B sampling plans, as listed in table XI shall be as follows:

Group B plan	AQL	Inspection level for normal inspection	Inspection level for reduced inspection
B-2	6.5%	L-7	L-5

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4.3.2.2 Disposition of sample units.- Sample units subjected to Group B test shall not be delivered on the contract or order.

4.3.2.3 Procedure in case of failure.- When an inspection lot is rejected, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

Table XI - Government Group B inspection

Inspection	Rqt. Par.	Test Par.	Group B plans
Low-temperature operation	3.16	4.4.8	B-2
High-temperature operation	3.17	4.4.9	B-2

4.3.3 Group C inspection.- Group C inspection shall be as specified in table XII.

4.3.3.1 Sampling for Group C inspection.- Samples for group C inspection shall be selected as follows:

(a) Sub-Group I Tests.- Regardless of the circuit configuration of rectifiers on contract or order, each month, basic bridge sample units assembled by the contractor from rectifier cells of the largest plate area selected from production sh be subjected to group A tests. Eight basic bridge sample units which have passed group A tests are required for group C tests in this sub-group. Four of these units shall be tested and the other four units shall remain in reserve to be tested in case of a failure in the first units tested. (See 4.3.3.3).

(b) Sub-Group II Tests.- Regardless of the circuit configuration of rectifiers on contract or order, not more than once every six months, basic bridge sample units assembled by the contractor from rectifier cells Style RS15, or the next largest style in production, shall be subjected to group A tests. Eight basic bridge sample units which have passed group A tests are required for group C tests in the sub-group. Four of these units shall be tested and the other four units shall remain in reserve to be tested in case of a failure in the first units tested. (See 4.3.3.3.)

4.3.3.2 Disposition of sample units.- Sample units subjected to group C tests shall not be delivered on the contract or order. Reserve sample units not tested will be returned to the contractor.

4.3.3.3 Non-compliance.- The following criteria shall be used to denote failure in group C tests:

(a) Failure of more than one sample unit in either sub-group I or sub-group II.

(b) Failure of a reserve sample unit.

when failure in group C tests occur, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and

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details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

Table XII - Government group C inspection

Test	Rqt. Par.	Test Par.
Subgroup I (4 samples)		
Low-temperature exposure	3.14	4.4.6
Moisture resistance	3.15	4.4.7
Subgroup II (4 samples)		
Life	3.18	4.4.10

4.4 Method of test.-

4.4.1 Visual and mechanical inspection.- Rectifiers shall be inspected to verify that the materials, design and construction, physical dimensions, marking and workmanship are in accordance with the applicable requirements.

4.4.2 Forward voltage drop.- Rectifiers shall be connected as shown in figure 3 and tested as specified in 4.4.2.1 or 4.4.2.2. (See 3.10).

4.4.2.1 Forward voltage drop (short circuit test).- The forward voltage drop shall be measured after 5 ± 1/2 minutes of operation at the specified rated forward current. The nominal voltage drop of the ammeter shall not be greater than 50 millivolts, and the voltage drop in the output circuit (output terminals of the rectifiers) shall not be greater than 100 millivolts.

4.4.2.2 Alternate forward voltage drop.- The forward voltage drop for group A testing shall be measured immediately (cold) after application of the specified rated forward current. To perform this test on half-wave rectifiers, one rectifier of the type to be tested shall be used as a load balance (R), as shown in figure 3. It is essential that the load balance have electrical characteristics similar to the rectifiers to be tested.

4.4.3 Reverse current.- Rectifiers shall be connected as shown in figure 3 and tested as specified in 4.4.3.1 or 4.4.3.2.

4.4.3.1 Reverse current (open-circuit test).- Reverse current shall be measured after 5 ± 1/2 minutes of operation at the specified rated reverse voltage. The rated reverse voltage shall be measured by means of an ac voltmeter of sufficiently high impedance to reduce error due to excessive current drain by the voltmeter or correction shall be made for the current drain of the voltmeter.

4.4.3.2 Alternate reverse current.- The reverse current for group A testing shall be measured within 5 minutes after application of the specified rated reverse voltage. To perform this test on half-wave rectifiers, one rectifier of the type to be tested shall be used as a load balance (R), as shown in figure 3. It is essential that the load balance have electrical characteristics similar to the rectifiers to be tested.

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4.4.4 Dielectric strength.- Rectifiers shall be subjected to a test potential as specified in table XIII, for a period of 1 minute for Preproduction tests and for not less than 5 seconds nor more than 1 minute for acceptance tests. The test potential shall be applied between all current-carrying parts connected together and the mounting members. Voltages above 1,000 volts rms shall be applied gradually at a rate not exceeding 500 volts rms per second; however, for acceptance tests, the test voltage may be applied instantaneously at the option of the manufacturer. (See 3.12)

Table XIII - Dielectric test voltages

Reverse voltage rating (volts rms.)	Test potential (volts rms)
60 and under	600
61 to 90, inclusive	900
Over 90	1,000 plus twice rated reverse voltage ratings (rms). but not to exceed 3000 volts

4.4.5 Insulation resistance.- The insulation resistance between all current-carrying parts connected together and the mounting members shall be measured using a 500-volt insulation tester. (See 3.13)

4.4.6 Low-temperature exposure.- Rectifiers shall be conditioned at $-65\ ^{+0}_{-3}$ °C for 2 hours. At the conclusion of this conditioning, the rectifiers shall be removed from the conditioning chamber and allowed to remain at room ambient temperature for at least 4 hours, after which the forward voltage drop specified in 4.4.2.1 shall be measured. (See 3.14)

4.4.7 Moisture resistance (see 3.15).- Rectifiers shall be subjected to the moisture-resistance test as specified in Method 106 of Standard MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting - on racks by normal mounting means.

(b) Initial measurements - not applicable.

(c) Polarization and load - not applicable.

(d) Final measurements - conditioning at 90-95 percent humidity not applicable. Rectifiers shall be tested for forward voltage drop, reverse current, dielectric strength, and insulation resistance. (See 4.4.2, 4.4.3, 4.4.4, and 4.4.5 respectively.)

4.4.8 Low-temperature operation.- Rectifiers shall be exposed to an ambient temperature of -55°C for one hour and then operated at rated current and voltages at $-55\ ^{+0}_{-3}$ °C for 2 hours. While at this ambient temperature, the forward voltage drop and reverse current shall be measured as specified in 4.4.2 and 4.4.3. (See 3.16).

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4.4.9 High-temperature operation. - Rectifiers shall be operated at an ambient temperature of 125 $\pm^{0}_{3}$ °C for a period of 4 hours. The input voltage shall be 85% of the rated voltage. The output current into a resistive load shall be adjusted to 10 percent of the specified rated forward current. Following this conditioning, and while at this elevated temperature, the reverse current shall be measured as specified in 4.4.3. (See 3.17).

4.4.10 Life. - Rectifiers shall be operated continuously at the specified rated reverse voltage and forward current into a resistive load, while maintained at an ambient temperature of 105°C for 500 hours. Forward voltage drop shall be measured at room temperature before and after test. (See 3.18).

4.4.11 Mechanical Shock. - Rectifiers shall be mounted in each of three (3) mutually perpendicular planes on the mounting adapter shown on figure 6E of specification MIL-S-901 and shall be subjected to the type C test of specification MIL-S-901. Following the test, the forward voltage drop and reverse current shall be measured and the dielectric strength test performed. (See 3.19).

4.4.12 Vibration (see 3.20). - Rectifiers shall be subjected to the vibration test specified in Method 201 of Standard MIL-STD-202. The following details and exception shall apply:

(a) Mounting - by normal mounting means.

(b) Initial measurements - not applicable.

(c) Motion - 1 hour in each of three mutually perpendicular directions (total of 3 hours).

(d) Final measurements - Rectifiers shall be visually examined for loosening of parts or evidence of mechanical damage, and tested for forward voltage drop, reverse current, and dielectric strength. (See 4.4.1, 4.4.2, 4.4.3 and 4.4.4 respectively.)

4.4.13 Corrosion (see 3.21). - Rectifiers shall be subjected to the salt-spray (corrosion) test as specified in Method 101 of Standard MIL-STD-202, condition B. The duration of this test shall be 50 hours. Rectifiers shall be visually inspected, and tested for forward voltage drop, reverse current and dielectric strength. (See 4.4.1, 4.4.2, 4.4.3, and 4.4.4 respectively.)

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. -

5.1.1 Level A - Military package. - Rectifiers, Metallic, Selenium (High Temperature) shall be preserved and packaged in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and described in the following paragraphs.

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5.1.1.1 Rectifiers, Metallic, Selenium, (High Temperature).- Rectifiers shall be packaged individually Method IA-8 as follows: Cushion each rectifier within a wrap of flexible, single-face, corrugated fiber board, Type I in accordance with Specification LLL-F-291. Secure the cushioning with gummed paper tape, Class 2 in accordance with Specification UU-T-111. Inclose each cushioned rectifier within a close-fitting bag fabricated of Water Vaporproof barrier material qualified for Class 2 application in conformance with the requirements of Specification MIL-B-131. Heat seal closure. Place each water-vapor-proofed rectifier within a second close-fitting, folding carton, Style II, Type A, Class A, conforming to Specification PPP-B-566. Seal box closure by means of gummed paper tape, conforming to Specification UU-T-111.

5.1.1.2 Intermediate package.- Place quantity of rectifiers having the same stock number, packaged as specified in 5.1.1.1, within a close-fitting, regular slotted style, Grade 5, Type CF, fiberboard box conforming to the requirements of Specification JAN-P-108. Fill all voids with flexible, single-face, corrugated fiberboard Type II conforming to Specification LLL-F-291. Seal closure with adhesive in accordance with the appendix of the box specification. In addition, entire box closure, corners and joints shall be sealed with water-resistant, pressure-sensitive tape Type I, Grade A. Olive Drab, conforming to Specification JAN-P-127.

5.1.2 Level B-Limited military package.- Rectifiers, Metallic, Selenium (High Temperature) shall be preserved and packaged in accordance with procedures specified in 5.1.1.

5.1.3 Level C - Minimum military package.- Rectifiers, Metallic Selenium (High Temperature) shall be packaged in accordance with the commercial practice and in a manner that will guarantee protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity at lowest transportation rate.

5.2 Packing.-

5.2.1 Level A - Military pack.- Place a quantity of Rectifiers, Metallic Selenium (High Temperature) in a Style 4 Nailed wood box conforming to Specification JAN-P-106. Fabricate the box to fit contents snugly. Approximate gross weight of any shipping container shall not exceed 150 pounds. Close the shipping container as specified in the appendix of the box specification.

5.2.1.1 Metal Strapping.- Shipping containers shall be strapped in conformance with the requirements of the appendix of the container specification only for direct shipment to ports.

5.2.2 Level B-Limited military pack.- Place a quantity of Rectifiers, Metallic, Selenium (High Temperature) in a Style 4, nailed wood box conforming to Specification NN-B-621. Fabricate the box to fit the contents snugly. Approximate gross weight of any shipping container shall not exceed 200 pounds. Closure shall be made in accordance with the box specification. Strapping is not required.

5.2.3 Level C-Minimum military pack.- Rectifiers, Metallic, Selenium (High Temperature) shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification for rail shipment, National Motor Freight

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Classification for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of transportation employed.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Military Standard MIL-STD-129.

6. NOTES

6.1 Intended Use.- Rectifiers covered by this specification are intended primarily for use in power rectification. For application in such items as magnetic amplifiers or as dc blocking components, additional tests other than those specified herein are necessary to determine the acceptability of the rectifiers.

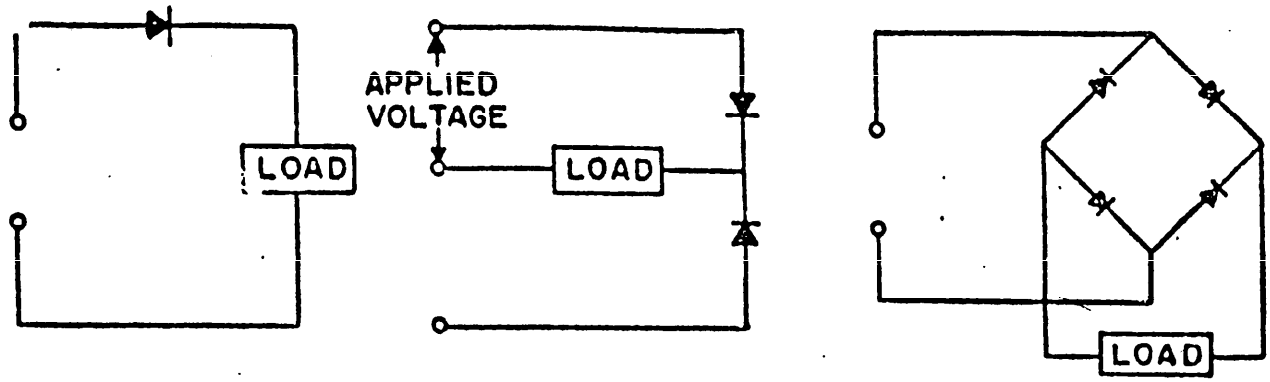
6.2 Ordering data.- Procurement documents should specify the title, number and date of this specification, and any amendment thereto, and the type designation of the rectifier required.

6.3 Electrical ratings of rectifiers included in this specification are listed in figures appended to this specification.

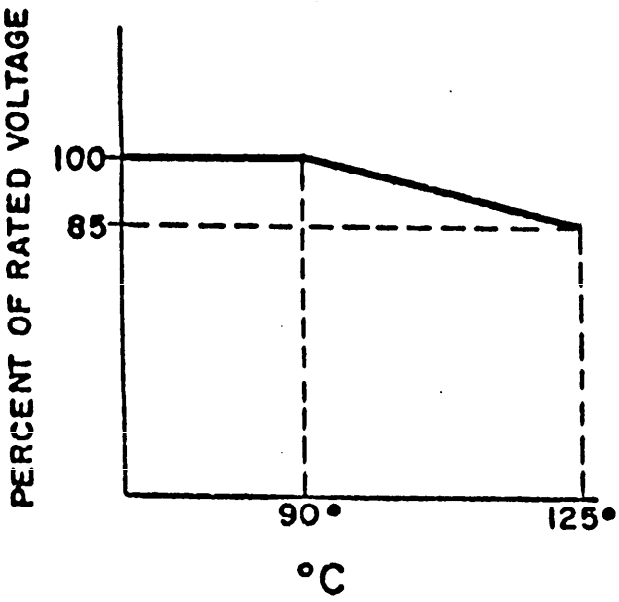
6.4 Mounting brackets.- Mounting brackets for normal applications are shown in figure 4.

6.4.1 High impact shock brackets.- Mounting brackets for high impact shock application are shown in figure 6E of Specification MIL-S-901.

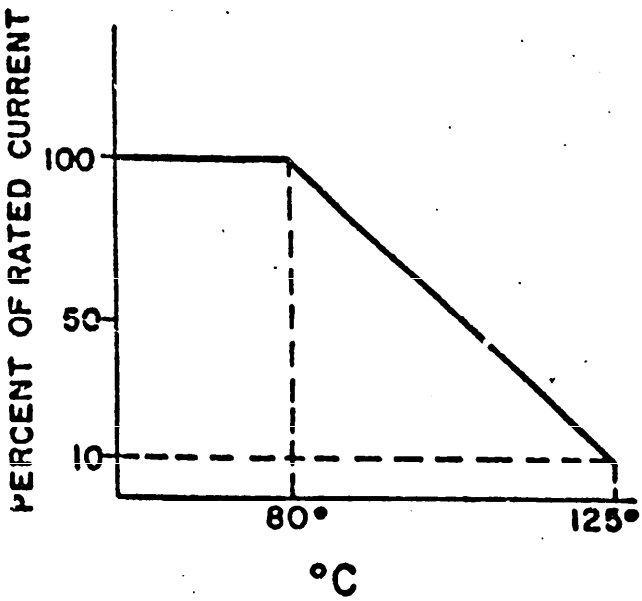
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CIRCUIT DIAGRAMS



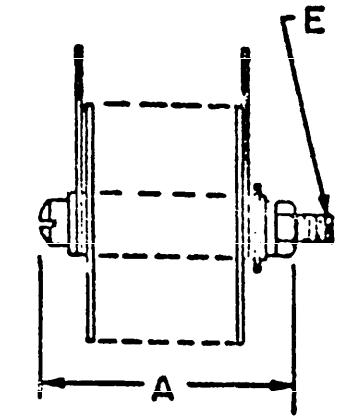
VOLTAGE DERATING CURVE



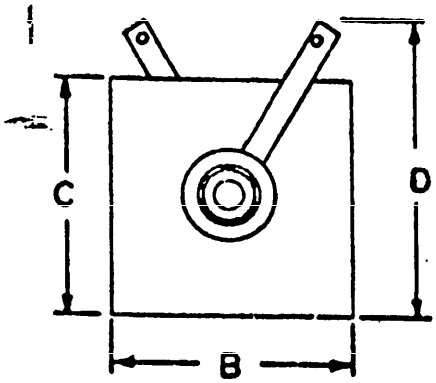
CURRENT DERATING CURVE

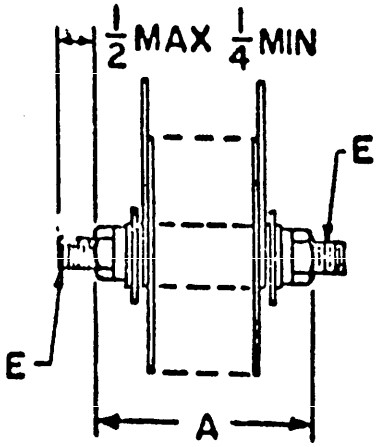
FIGURE 1.

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BOLT TYPE MOUNTING

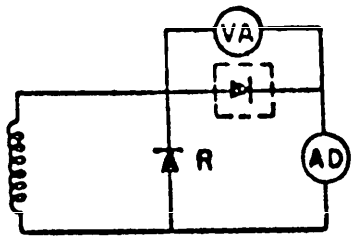
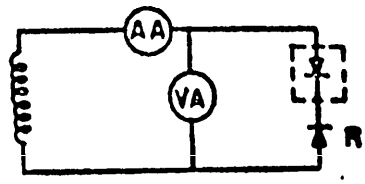
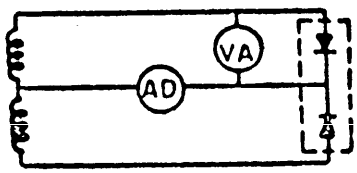
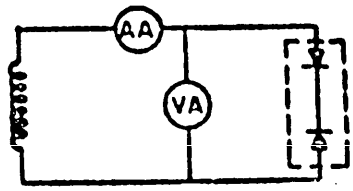
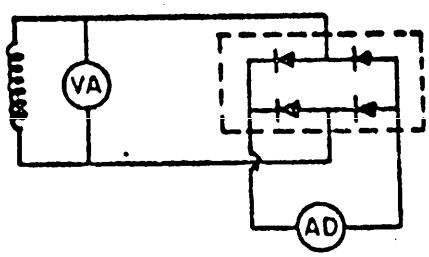
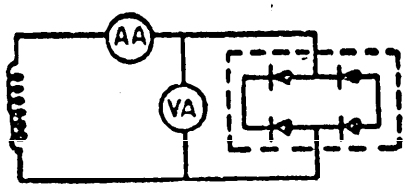




STUD TYPE MOUNTING

FIGURE 2

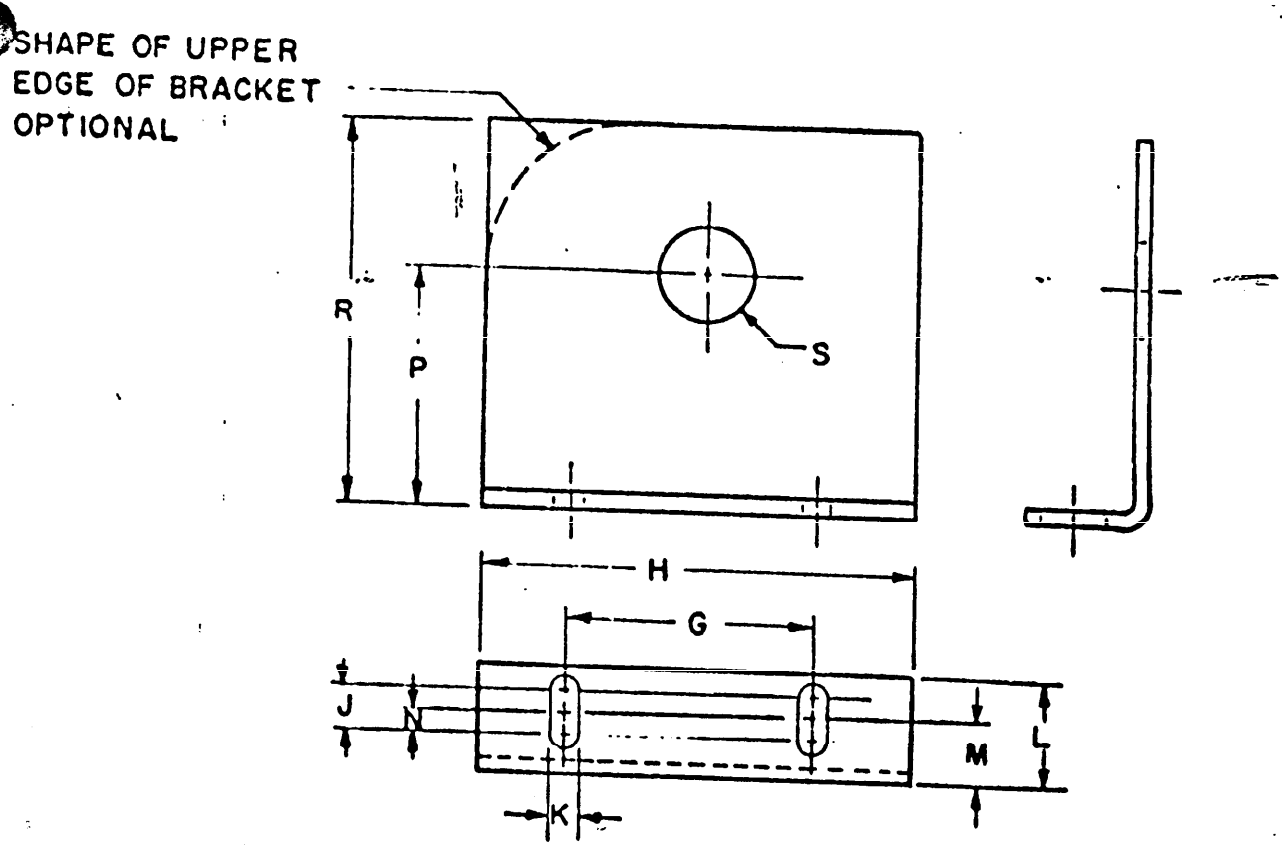
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	Forward voltage drop	Reverse current
a. Half Wave (HS)		
b. Center Tap (CS)		
c. Bridge (BS) Doubler (DS)		

AD - dc ammeter, permanent magnet, moving coil type
AA - dc ammeter or milliammeter, rms reading
VD - dc voltmeter
VA - ac voltmeter, VTVM
R - load balance

Figure 3. Test circuits for forward voltage drop and reverse current, single phase.

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Style	G	H 1/	J	K	L	M	N	P	R 2/	Mounting Screw	S 3/
RS15	9/16		3/16	5/32	5/8	5/16	3/32	1-1/16		6-32	8-32
RS25	9/16		3/16	5/32	5/8	5/16	3/32	1-3/16		6-32	10-32
RS35	9/16		3/16	5/32	5/8	5/16	3/32	1-3/8		6-32	10-32
RS45	1-1/8		1/4	7/32	7/8	1/2	1/8	1-3/4		10-32	1/4-20
RS55	1-1/2		9/32	9/32	1	19/32	9/64	2-3/16		1/4-20	1/4-20
RS65	1-1/2		9/32	9/32	1	19/32	9/64	2-11/16		1/4-20	5/16-18
RS75	2-1/2		9/32	9/32	1	19/32	9/64	3-1/2		1/4-20	5/16-18
RS85	2-1/2		9/32	9/32	1	19/32	9/64	4		1/4-20	5/16-18

1/ Shall not exceed the width of the cell.

2/ Shall not extend beyond the top of the rectifier cell.

3/ Sufficient clearance for mounting screw.

Figure 4. Mounting brackets for normal applications

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Single Phase Half Wave Style RS15
Thread Size E, #8 -32 NF-2A
Cell width B - 1.050" Max
Cell height C - 1.050" Max
Overall height D - 1.425" Max

Type Designation 1/	DC Output Ratings 2/		Length A 3/	Forward Voltage Drop, Maximum	AC Applied Voltage Maximum (RMS)	Number of Cells per Rectifier, Minimum
	Current	Voltage				
	Ampere	Volts	Inches	Volts	Volts	
RS15-100HS141	.140	10	11/16	2.5	26	1
RS15-200HS141	.140	20	7/8	5.0	52	2
RS15-300HS141	.140	30	1-1/6	7.5	78	3
RS15-400HS141	.140	40	1-1/4	10.0	104	4
RS15-500HS141	.140	50	1-7/16	12.5	130	5
RS15-600HS141	.140	60	1-19/32	15.0	156	6
RS15-800HS141	.140	80	1-15/16	20.0	208	8
RS15-101HS101	.100	100	2-7/32	18.8	260	10
RS15-101HS141	.140	100	2-9/16	25.0	260	10
RS15-121HS800	.080	120	2-1/2	22.5	312	12
RS15-121HS101	.100	120	3-3/16	30.0	312	12
RS15-141JS800	.080	140	2-11/16	26.3	364	14
RS15-141HS101	.100	140	3-19/32	35.0	364	14
RS15-161HS800	.080	160	3	30.0	416	16
RS15-161HS101	.100	160	4	40.0	416	16
RS15-181HS800	.080	180	3-9/32	33.8	468	18
RS15-181HS101	.100	180	4-13/32	45.0	468	18
RS15-201HS800	.080	200	3-9/16	37.5	520	20
RS15-201HS101	.100	200	4-13/16	50.0	520	20
RS15-231HS800	.080	230	3-31/32	43.2	598	23
RS15-231HS101	.100	230	5-13/32	57.5	598	23
RS15-261HS600	.060	260	4-11/32	48.8	676	26
RS15-261HS800	.080	260	6-1/32	65.0	676	26
RS15-301HS600	.060	300	4-31/32	56.3	780	30
RS15-351HS600	.060	350	5-5/8	65.7	910	35
RS15-401HS600	.060	400	6-3/8	75.0	1040	40

1/ Insert appropriate letter for mounting method. Bolt (letter B) is available only for rectifiers with a length A dimension of 1-3/4 inches or less. Stud (letter S) is available for rectifiers of all lengths.

2/ Ratings are for resistive and inductive loads and continuous operation at 80°C ambient temperature. For capacitive or battery loads, the ratings shall be 80 percent of those specified. Ratings at higher ambients shall be determined from the derating curves of Figure 1.

3/ For length A dimensions less than 6 inches, tolerance is ±1/16 inch; from 6 to 12 inches, inclusive, tolerance is ±1/8 inch; and for 12 inches and greater, tolerance is ±1/4 inch.

Fig. 5

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SINGLE PHASE CENTER TAP STYLE RS15

Thread size E #8-32 NF-2A — Cell height C - 1.050" Max
Cell width B - 1.050" Max — Overall height D - 1.425" Max

Type Designation 1/	DC Output Ratings 2/		Length A 3/	Forward Voltage Drop, Maximum	AC Applied Voltage Maximum (RMS)	Number of Cells per Rectifier, Minimum
	Current	Voltage				
	Ampere	Volts	Inches	Volts	Volts	
RS15-100CS281	.280	10	15/16	2.5	13	2
RS15-200CS281	.280	20	1-5/16	5.0	26	4
RS15-300CS281	.280	30	1-21/32	7.5	39	6
RS15-400CS281	.280	40	2	10.0	52	8
RS15-500CS201	.200	50	2-9/32	9.4	65	10
RS15-500CS281	.280	50	2-5/8	12.5	65	10
RS15-600CS161	.160	60	2-9/16	11.3	78	12
RS15-600CS221	.220	60	3-1/4	15.0	78	12
RS15-800CS161	.160	80	3-1/16	15.0	104	16
RS15-800CS221	.220	80	4-1/16	20.0	104	16
RS15-101CS161	.160	100	3-5/8	18.8	130	20
RS15-101CS221	.220	100	4-7/8	25.0	130	20
RS15-121CS161	.160	120	4-3/16	22.5	156	24
RS15-121CS221	.220	120	5-11/16	30.0	156	24
RS15-141CS121	.120	140	4-3/4	26.3	182	28
RS15-141CS161	.160	140	6-1/2	35.0	182	28
RS15-161CS121	.120	160	5-5/16	30.0	208	32
RS15-181CS121	.120	180	5-7/8	33.8	234	36
RS15-201CS121	.120	200	6-7/16	37.5	260	40

1/ Insert appropriate letter for mounting method. Bolt (letter B) is available only for rectifiers with a length A dimension of 1-3/4 inches or less. Stud (letter S) is available for rectifiers of all lengths.

2/ Ratings are for resistive and inductive loads and continuous operation at 80°C ambient temperature. For capacitive or battery loads, the ratings shall be 80 percent of those specified. Ratings at higher ambients shall be determined from the derating curves of Figure 1.

3/ For length A dimensions less than 6 inches, tolerance is ±1/16 inch; from 6 to 12 inches, inclusive, tolerance is ±1/8 inch; and for 12 inches and greater, tolerance is ±1/4 inch.

Fig. 6.

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SINGLE PHASE, BRIDGE, STYLE RS15

Thread size E, #8 - 32 NF - 2A
Cell width B - 1.050" Max
Cell Height C - 1.050" Max
Overall height D - 1.425" Max

Type Designation 1/	DC Output Ratings 2/		Length A 3/	Forward Voltage Drop, Maximum	AC Applied Voltage Maximum (RMS)	Number of Cells per Rectifier, Minimum
	Current	Voltage				
	Ampere	Volts	Inches	Volts	Volts	
RS15-100BS281	.280	10	1-3/8	5.0	18	4
RS15-200BS281	.280	20	1-3/8	5.0	26	4
RS15-400BS281	.280	40	2-1/16	10.0	52	8
RS15-600BS161	.160	60	2-5/8	11.3	78	12
RS15-600BS221	.220	60	3-5/16	15.0	78	12
RS15-800BS161	.160	80	3-1/8	15.0	104	16
RS15-800BS221	.220	80	4-1/8	20.0	104	16
RS15-101BS161	.160	100	3-11/16	18.8	130	20
RS15-101BS221	.220	100	4-15/16	25.0	130	20
RS15-121BS161	.160	120	4-1/4	22.5	156	24
RS15-121BS221	.220	120	5-3/4	30.0	156	24
RS15-141BS121	.120	140	4-13/16	26.3	182	28
RS15-141BS161	.160	140	6-9/16	35.0	182	28
RS15-161BS121	.120	160	5-3/8	30.0	208	32
RS15-181BS121	.120	180	5-15/16	33.8	234	36
RS15-201BS121	.120	200	6-1/2	37.5	260	40

1/ Insert appropriate letter for mounting method. Bolt (letter B) is available only for rectifiers with a length A dimension of 1-3/4 inches or less. Stud (letter S) is available for rectifiers of all lengths.

2/ Ratings are for resistive and inductive loads and continuous operation at 80°C ambient temperature. For capacitive or battery loads, the ratings shall be 80 percent of those specified. Ratings at higher ambients shall be determined from the derating curves of Figure 1.

3/ For length A dimensions less than 6 inches, tolerance is ±1/16 inch; from 6 to 12 inches, inclusive, tolerance is ±1/8 inch; and for 12 inches and greater, tolerance is ±1/4 inch.

Fig. 7

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SINGLE PHASE, HALF WAVE, STYLE RS25

Thread size E, #10-32 NF-2A — Cell height C - 1.300" Max

Cell Width B - 1.300" Max — Overall height D - 1.675" Max

Type Designation 1/	DC Output Ratings 2/		Length A3/	Forward Voltage Drop, Maximum	AC Applied Voltage Maximum (RMS)	Number of Cells per Rectifier, Minimum
	Current	Voltage				
	Ampere	Volts	Inches	Volts	Volts	
RS25-100HS281	.280	10	3/4	2.5	26	1
RS25-200HS281	.280	20	15/16	5.0	52	2
RS25-300HS281	.280	30	1-1/8	7.5	78	3
RS25-400HS281	.280	40	1-5/16	10.0	104	4
RS25-500HS281	.280	50	1-1/2	12.5	130	5
RS25-600HS281	.280	60	1-11/16	15.0	156	6
RS25-800HS281	.280	80	2-1/32	20.0	208	8
RS25-101HS201	.200	100	2-11/32	18.8	260	10
RS25-101HS281	.280	100	2-3/4	25.0	260	10
RS25-121HS161	.160	120	2-21/32	22.5	312	12
RS25-121HS201	.200	120	3-7/16	30.0	312	12
RS25-141HS161	.160	140	2-31/32	26.3	364	14
RS25-141HS201	.200	140	3-29/32	35.0	364	14
RS25-161HS161	.160	160	3-1/4	30.0	416	16
RS25-161HS201	.200	160	4-11/32	40.0	416	16
RS25-181HS161	.160	180	3-9/16	33.8	468	18
RS25-181HS201	.200	180	4-13/16	45.0	468	18
RS25-201HS161	.160	200	3-27/32	37.5	520	20
RS25-201HS201	.200	200	5-1/4	50.0	520	20
RS25-231HS161	.160	230	4-9/32	43.2	598	23
RS25-231HS201	.200	230	5-15/16	57.5	598	23
RS25-261HS121	.120	260	4-25/32	48.8	676	26
RS25-261HS161	.160	260	6-15/32	65.0	676	26
RS25-301HS121	.120	300	5-3/8	56.3	780	30
RS25-351HS121	.120	350	6-5/32	65.7	910	35
RS25-401HS121	.120	400	6-29/32	75.0	1040	40

1/ Insert appropriate letter for mounting method. Bolt (letter B) is available only for rectifiers with a length A dimension of 1-3/4 inches or less. Stud (letter S) is available for rectifiers of all lengths.

2/ Ratings are for resistive and inductive loads and continuous operation at 80°C ambient temperature. For capacitive or battery loads, the ratings shall be 80 percent of those specified. Ratings at higher ambients shall be determined from the derating curves of Figure 1.

3/ For length A dimensions less than 6 inches, tolerance is ±1/16 inch; from 6 to 12 inches, inclusive, tolerance is ±1/8 inch; and for 12 inches and greater, tolerance is ±1/4 inch.

Fig 8

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SINGLE PHASE, CENTER TAP STYLE RS25

Thread size E, #10-32 NF-2A
Cell Width B-1.300" Max
Cell height C-1.300" Max
Overall height D-1.675" Max

Type Designation 1/	DC Output Ratings 2/		Length A3/	Forward Voltage Drop, Maximum	AC Applied Voltage, Maximum (RMS)	Number of Cells per Rectifier, Minimum
	Current	Voltage				
	Ampere	Volts	Inches	Volts	Volts	
RS25-100CS561	.560	10	1	2.5	13	2
RS25-200CS561	.560	20	1-3/8	5.0	26	4
RS25-300CS561	.560	30	1-3/4	7.5	39	6
RS25-400CS561	.560	40	2-3/32	10.0	52	8
RS25-500CS401	.400	50	2-13/32	9.4	65	10
RS25-500CS561	.560	50	2-13/16	12.5	65	10
RS25-600CS321	.320	60	2-23/32	11.3	78	12
RS25-600CS441	.440	60	3-1/2	15.0	78	12
RS25-800CS321	.320	80	3-5/16	15.0	104	16
RS25-800CS441	.440	80	4-13/32	20.0	104	16
RS25-101CS321	.320	100	3-29/32	18.8	130	20
RS25-101CS441	.440	100	5-5/16	25.0	130	20
RS25-121CS321	.320	120	4-17/32	22.5	156	24
RS25-121CS441	.440	120	6-7/32	30.0	156	24
RS25-141CS241	.240	140	5-1/8	26.3	182	28
RS25-141CS321	.320	140	7-1/8	35.0	182	28
RS25-161CS241	.240	160	5-3/4	30.0	208	32
RS25-181CS241	.240	180	6-3/8	33.8	234	36
RS25-201CS241	.240	200	6-31/32	37.5	260	40

1/ Insert appropriate letter for mounting method. Bolt (letter B) is available only for rectifiers with a length A dimension of 1-3/4 inches or less. Stud (letter S) is available for rectifiers of all lengths.

2/ Ratings are for resistive and inductive loads and continuous operation at 80°C ambient temperature. For capacitive or battery loads, the ratings shall be 80 percent of those specified. Ratings at higher ambients shall be determined from the derating curves of Figure 1.

3/ For length A dimensions less than 6 inches, tolerance is ±1/16 inch; from 6 to 12 inches, inclusive, tolerance is ±1/8 inch; and for 12 inches and greater, tolerance is ±1/4 inch.

Fig 9

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SINGLE PHASE, BRIDGE, STYLE RS25

Thread size E. #10-32 NF-2A
Cell width B - 1.300" Max
Cell height C - 1.300" Max
Overall height D - 1.675" Max

Type Designation 1/	DC Output Ratings 2/		Length A 3/	Forward Voltage Drop, Maximum	AC Applied Voltage, Maximum (RMS)	Number of Cells per Rectifier, Minimum
	Current	Voltage				
	Ampere	Volts	Inches	Volts	Volts	
RS25-100BS561	.560	10	1-7/16	5.0	18	4
RS25-200BS561	.560	20	1-7/16	5.0	26	4
RS25-400BS561	.560	40	2-5/32	10.0	52	8
RS25-600BS321	.320	60	2-25/32	11.3	78	12
RS25-600BS441	.440	60	3-9/16	15.0	78	12
RS25-800BS321	.320	80	3-3/8	15.0	104	16
RS25-800BS441	.440	80	4-15/32	20.0	104	16
RS25-101BS321	.320	100	3-31/32	18.8	130	20
RS25-101BS441	.440	100	5-3/8	25.0	130	20
RS25-121BS321	.320	120	4-19/32	22.5	156	24
RS25-121BS441	.440	120	6-9/32	30.0	156	24
RS25-141BS241	.240	140	5-3/16	26.3	182	28
RS25-141BS321	.320	140	7-3/16	35.0	182	28
RS25-161BS241	.240	160	5-13/16	30.0	208	32
RS25-181BS241	.240	180	6-7/16	33.8	234	36
RS25-201BS241	.240	200	7-1/32	37.5	260	40

1/ Insert appropriate letter for mounting method. Bolt (letter B) is available only for rectifiers with a length A dimension of 1-3/4 inches or less. Stud (letter S) is available for rectifiers of all lengths.

2/ Ratings are for resistive and inductive loads and continuous operation at 80°C ambient temperature. For capacitive or battery loads, the ratings shall be 80 percent of those specified. Ratings at higher ambients shall be determined from the derating curves of Figure 1.

3/ For length A dimensions less than 6 inches, tolerance is ±1/16 inch; from 6 to 12 inches, inclusive, tolerance is ±1/8 inch; and for 12 inches and greater, tolerance is ±1/4 inch.

Fig 10

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SINGLE PHASE, HALF WAVE, STYLE RS35

Thread Size E-#10-32 NF-2A
Cell Width B - 1.650" Max
Cell Height C - 1.650" Max
Overall Height D - 2.025" Max

Type Designation 1/	DC Output Ratings 2/		Length A 3/	Forward Voltage Drop, Maximum	AC Applied Voltage, Maximum (RMS)	Number of Cells per Rectifier, Minimum
	Current	Voltage				
	Ampere	Volts	Inches	Volts	Volts	
RS35-100HS561	.560	10	13/16	2.5	26	1
RS35-200HS561	.560	20	1	5.0	52	2
RS35-300HS561	.560	30	1-3/16	7.5	78	3
RS35-400HS561	.560	40	1-3/8	10.0	104	4
RS35-500HS561	.560	50	1-15/32	12.5	130	5
RS35-600HS561	.560	60	1-3/4	15.0	156	6
RS35-800HS561	.560	80	2-1/8	20.0	208	8
RS35-101HS401	.400	100	2-15/32	18.8	260	10
RS35-101HS561	.560	100	2-27/32	25.0	260	10
RS35-121HS321	.320	120	2-13/16	22.5	312	12
RS35-121HS441	.440	120	3-11/16	30.0	312	12
RS35-141HS321	.320	140	3-5/32	26.3	364	14
RS35-141HS441	.440	140	4-1/8	35.0	364	14
RS35-161HS321	.320	160	3-1/2	30.0	416	16
RS35-161HS441	.440	160	4-11/16	40.0	416	16
RS35-181HS321	.320	180	3-13/16	33.8	468	18
RS35-181HS441	.440	180	5-3/16	45.0	468	18
RS35-201HS321	.320	200	4-1/8	37.5	520	20
RS35-201HS441	.440	200	5-11/16	50.0	520	20
RS35-231HS321	.320	230	4-21/32	43.2	598	23
RS35-231HS441	.440	230	6-7/16	57.5	598	23
RS35-261HS241	.240	260	5-1/8	48.8	676	26
RS35-261HS321	.320	260	7-3/16	65.0	676	26
RS35-301HS241	.240	300	5-25/32	56.3	780	30
RS35-351HS241	.240	350	6-5/8	65.7	910	35
RS35-401HS241	.240	400	7-7/16	75.0	1040	40

1/ Insert appropriate letter for mounting method. Bolt (letter B) is available only for rectifiers with a length A dimension of 1-3/4 inches or less. Stud (letter S) is available for rectifiers of all lengths.

2/ Ratings are for resistive and inductive loads and continuous operation at 80°C ambient temperature. For capacitive or battery loads, the ratings shall be 80 percent of those specified. Ratings at higher ambients shall be determined from the derating curves of Figure 1.

3/ For length A dimensions less than 6 inches, tolerance is ±1/16 inch; from 6 to 12 inches, inclusive, tolerance is ±1/8 inch; and for 12 inches and greater, tolerance is ±1/4 inch.

Fig 11

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SINGLE PHASE, CENTER TAP, STYLE RS35

Thread Size E - #10-32 NF-2A
Cell Width B - 1.650" Max
Cell Height C - 1.650" Max
Overall Height D - 2.025" Max

Type Designation 1/	DC Output Ratings 2/		Length A 3/	Forward Voltage Drop, Maximum	AC Applied Voltage, Maximum (RMS)	Number of Cells per Rectifier, Minimum
	Current	Voltage				
	Ampere	Volts	Inches	Volts	Volts	
RS35-100CS112	1.100	10	1-1/16	2.5	13	2
RS35-200CS112	1.100	20	1-7/16	5.0	26	4
RS35-300CS112	1.100	30	1-13/16	7.5	39	6
RS35-400CS112	1.100	40	2-3/16	10.0	52	8
RS35-500CS801	.800	50	2-17/32	9.4	65	10
RS35-500CS112	1.100	50	2-29/32	12.5	65	10
RS35-699CS641	.640	60	2-7/8	11.3	78	12
RS35-600CS881	.880	60	3-5/8	15.0	78	12
RS35-800CS641	.640	80	3-9/16	15.0	104	16
RS35-800CS881	.880	80	4-3/4	20.0	104	16
RS35-101CS641	.640	100	4-3/16	18.8	130	20
RS35-101CS881	.880	100	5-3/4	25.0	130	20
RS35-121CS641	.640	120	4-7/8	22.5	156	24
RS35-121CS881	.880	120	6-3/4	30.0	156	24
RS35-141CS481	.480	140	5-1/2	26.3	182	28
RS35-141CS641	.640	140	7-3/4	35.0	182	28
RS35-161CS881	.880	160	6-3/16	30.0	208	32
RS35-181CS481	.480	180	6-7/8	33.8	234	36
RS35-201CS481	.480	200	7-1/2	37.5	260	40

1/ Insert appropriate letter for mounting method. Bolt (letter B) is available only for rectifiers with a length A dimension of 1-3/4 inches or less. Stud (letter S) is available for rectifiers of all lengths.

2/ Ratings are for resistive and inductive loads and continuous operation at 80°C ambient temperature. For capacitive or battery loads, the ratings shall be 80 percent of those specified. Ratings at higher ambients shall be determined from the derating curves of Figure 1.

3/ For length A dimensions less than 6 inches, tolerance is ±1/16 inch; from 6 to 12 inches, inclusive, tolerance is ±1/8 inch; and for 12 inches and greater, tolerance is ±1/4 inch.

Fig 12

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a. Paragraph Number and Wording:

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